

CS.50600 · GRADUATE · 3 CREDITS

Database Systems

How database systems work, why they are built that way, and how recent research is rebuilding them.

INSTRUCTOR Kyoungmin Kim	EMAIL TBD	CLASS TIME (CLASSROOM) Tue / Thu 16:00–17:30 (E3-1 2445)
OFFICE HOURS The 17:30–18:30, or by appointment	L:L:C (AU) 3 : 0 : 3 (0)	EXAM TIME (CLASSROOM) No mid/final term, but 2 quizzes
PREREQUISITES Data structure	LANGUAGE English	EDUCATION 4.0 Q Y — Lecture 60% · Student-led 40%

01 Course Description

This course takes you inside a database system. We begin with a fast, three-week review of the undergraduate material — the relational model, storage, and indexing — then descend into the parts that undergraduate courses only wave at: how queries are executed and, above all, how they are **optimized**. From there we study how modern systems actually run analytics — vectorized engines like DuckDB, cloud warehouses like Snowflake, open formats like Parquet — and finish at the research frontier where machine learning and databases meet.

By the end, you should read a query plan the way a mechanic reads an engine, understand how data size and distribution change the right design, and critically evaluate — and begin to produce — database systems research.

02 Textbooks & Readings

- **Textbook:** Silberschatz, Korth & Sudarshan, *Database System Concepts*, 7th ed., McGraw-Hill, 2020.
- **Additional reading:** Bailis, Hellerstein & Stonebraker (eds.), *Readings in Database Systems* ("the Red Book"), 5th ed. — free at redbook.io.
- **Systems overview:** Hellerstein, Stonebraker & Hamilton, *Architecture of a Database System*, Foundations & Trends in Databases, 2007.
- **Research papers:** Recommended papers for weekly topics; either representative or state-of-the-art

There is no required purchase: the Red Book and all assigned papers are freely available online.

03 Grading

PROJECT 60%	REVIEW 10%	QUIZ 30%
-------------	------------	----------

- **Project 60%** — semester-long, teams of 1–2; any relevant topics in data management/processing, so you may pick a data management/processing issue/bottleneck in AI/HCI/HW/etc.; ultimate goal is to extract a research idea out of this project
 - Proposal presentation — 10%
 - Final presentation — 30%
 - Final report and deliverables — 20%
- **Paper review 10%** — short written review of 1 paper (select among listed candidates) including problem definition, motivation, methodology, and result
- **Quiz 30%** — 2 quizzes, each with 15%

04 Weekly Schedule

WK	TOPIC	DETAILS & KEY READINGS
PART I – FOUNDATIONS, QUICKLY (REVIEW)		
01	Introduction & history; relational model recap	What this course is; 50 years of database systems. Red Book Ch. 1.
02	Storage & indexing review	Pages, buffers, B+-trees, LSM-trees. DSC Ch. 13–14 (self-study + recap).
03	Query execution model	Operators, iterators, materialization vs. pipelining.
04	Join algorithms	Hash, sort-merge, index joins. Quiz 1 — SQL, plans, physical operators, execution.
PART II – QUERY OPTIMIZATION, IN DEPTH		
05	Query optimization I: the plan space	Selinger-style dynamic programming, join ordering, rewrite rules.
06	Query optimization II: cardinality & cost	Histograms, sampling, sketches; "How good are query optimizers, really?"
07	Adaptive & robust query processing	Re-optimization, robust plans, handling estimation error at runtime.
08	Mid-term: project proposal presentations	In class (midterm week — no exam). Proposal slides + written proposal due.
PART III – MODERN ANALYTICS ENGINES		
09	Vectorized OLAP engines: DuckDB & friends	Columnar layout, vectorized execution, morsel-driven parallelism. MonetDB/X100 (CIDR'05); DuckDB (SIGMOD'19).
10	Cloud warehouses & open formats	Snowflake architecture; Parquet, min/max zone maps, pruning & pushdown. Paper review due (Snowflake SIGMOD'16).
PART IV – THE REST OF THE ENGINE		
11	Transactions, concurrency & recovery	Serializability, MVCC, ARIES — compressed tour through modern engines.
12	Distributed & cloud-native processing	Partitioning, distributed joins, disaggregation, HTAP. Quiz 2 — optimization & OLAP internals.
PART V – RESEARCH FRONTIER: DATABASES MEET AI		
13	Learned database components	Learned indexes, learned cardinality estimation & optimizers.
14	Vector data management & retrieval for AI	ANN indexes (HNSW, IVF), vector/hybrid search, RAG infrastructure.
15	LLMs and databases	NL2SQL, table reasoning, agentic workloads; open problems.
16	Final project presentations	Presentations & demos. Report due end of week.

05 Course Project

The heart of the course. In teams of one or two, you will either **build** a nontrivial database system component (e.g., a join operator, a simple optimizer, a vector index) and evaluate it honestly, or **investigate** a research question with a rigorous experimental study. Strong projects have grown into workshop and conference papers — aim for that. Checkpoints: proposal presentation (Week 8), final presentation & report (Week 16), with informal check-ins in office hours in between.

06 Policies

- **Late policy:** Late submissions are not permitted without prior permission.
- **Collaboration:** Discuss freely, but write alone for reviews, which must be your own words. Project is team-based.
- **AI tools:** Permitted as assistants (coding, editing) with disclosure in your report. You are responsible for correctness, and quizzes are closed-AI.